

Qualifying Exam in Microeconomics
Arizona State University
June 2013

There are five questions and you have four hours to complete the exam.

1. Consider a two-period game with three players, an Owner of a firm, P , and two potential Managers, M_1 and M_2 . The firm's total profit at each date is normalized to 1. At each date, P first chooses one of the managers to run the firm. The manager chosen at date t then decides what fraction $x_t \in [0, 1]$ of that date's profit to expropriate. If manager i is chosen and expropriates $x_t^i \in [0, 1]$ at date t , then manager i gets x_t^i dollars, P gets $1 - x_t^i$ dollars and manager $j \neq i$ gets 0 dollars at date t . A player's preferences over outcomes of the game are represented by the player's discounted monetary payoff; the common discount factor is $\delta \in (0, 1)$. Before choosing date-2 actions, every player observes the date-1 history (which manager i is chosen, and the fraction x_1^i of profit the chosen manager i expropriates). In what follows confine your attention to pure strategies.
 - (a) Describe the strategy sets for players P and M_1 in this game. Give an example of a strategy for P and for M_1 .
 - (b) What is the highest discounted payoff that player P can get in a subgame-perfect Nash equilibrium (SPNE)? Identify a subgame perfect strategy profile that supports this payoff, and explain clearly why it is subgame perfect and why player P cannot get a higher payoff.
 - (c) Are there Nash equilibria of this game that are not subgame perfect? If so, identify one; if not explain why all Nash equilibria are subgame perfect.
2. A firm sells two goods, 1 and 2. If it sells $q_i \geq 0$ units of good i for $i = 1, 2$, then its profit is $\phi(q_1, q_2, \theta) = R_1(q_1, \theta) + R_2(q_2) - C(q_1, q_2)$, where R_i is revenue from good i , $\theta \in [0, 1]$ is a parameter affecting the revenue for good 1 and $C(\cdot)$ is the firm's cost function. Assume throughout that R_1 satisfies *strictly increasing differences* in q_1 and θ .
 - (a) For $\theta \in \{0, 1\}$ let $q^\theta = (q_1^\theta, q_2^\theta)$ maximize the firm's profit at θ . Does it follow that $q_1^1 \geq q_1^0$? If so, prove it, if not give sufficient conditions for the conclusion and explain why the conditions you identify are sufficient.
 - (b) Using the same notation from part (a), does it follow that $(q_1^1, q_2^1) \geq (q_1^0, q_2^0)$? If so, prove it, if not give sufficient conditions for the conclusion and explain why the conditions you identify are sufficient.
 - (c) For this part q_2 is fixed, so it is without loss of generality to suppress it and set $\phi(q, \theta) = R(q, \theta) - C(q)$, where q the output of good 1 and R is now revenue for good 1 (and so satisfies strictly increasing differences). Suppose now that θ is unknown to the firm. For $t = 0, 1$, let $f(\cdot, t)$ be a continuous density function which is positive on $[0, 1]$ and for $t = 0, 1$, let q^t maximize the firm's expected profit when its beliefs about θ are given by the density $f(\cdot, t)$. Suppose that the likelihood ratio $f(\theta, 1)/f(\theta, 0)$ is strictly increasing in θ on $[0, 1]$ and suppose that q^t maximizes expected profit when the firm's beliefs are given by the density $f(\cdot, t)$ for $t = 0, 1$. Prove that that $q^1 \geq q^0$.

3. In what follows $\{(P_t, Q_{1t}, \dots, Q_{nt})\}_{t=1}^T$ is a finite sequence of $T \geq 2$ price and quantity observations from an industry with $n \geq 2$ firms selling identical products. (Firm i 's quantity at observation t is Q_{it} .) Each firm i produces a single output with technology represented by a cost function c_i that is *linear* (that is, the technology is constant returns). The cost functions do not vary with t , but the demand for the good possibly does. Neither the cost functions nor the demand functions are observed. Each price-quantity observation is *positive*: $(P_t, Q_{1t}, \dots, Q_{nt}) \gg 0$ for $t = 1, \dots, T$.

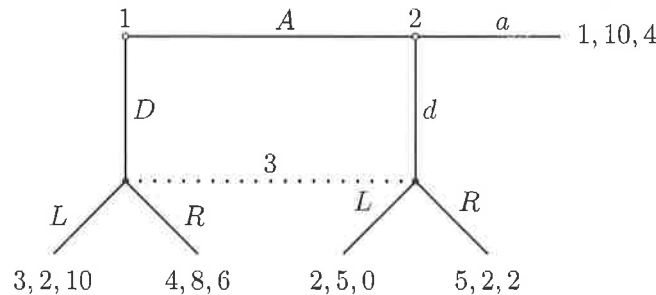
- (a) Suppose that for $t = 1, \dots, T$, the quantity observation $(Q_{1t}, \dots, Q_{nt})$ is a Nash equilibrium of a Cournot Game with inverse demand $\tilde{P}_t(\cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ satisfying the usual regularity conditions: for each t , $\tilde{P}_t(\cdot)$ is nonincreasing on $\mathbb{R}_+$; $\tilde{P}_t(0) > 0$; and on $\{Q \in \mathbb{R}_+ \mid \tilde{P}_t(Q) > 0\}$ it is C^1 with $\tilde{P}'_t < 0$. Show that there exist n positive numbers $\gamma_1, \dots, \gamma_n$ such that

$$\frac{P_t - \gamma_i}{Q_{it}} = \frac{P_t - \gamma_j}{Q_{jt}} > 0 \quad (1)$$

for every firm i and j and every $t = 1, \dots, T$, where $P_t = \tilde{P}_t(\sum_i Q_{it})$. Give an interpretation of the numbers γ_i .

- (b) Maintaining the assumptions from part (a), suppose that firm 1's market share—its output as a fraction of the industry's output—is (strictly) higher than any other firm at $t = 1$, and that its market share is *strictly increasing* in t . Prove that the price must *decrease* with t . (It follows that the hypothesis in part (a) is *testable*.)
- (c) Show that *any* positive sequence $\{(P_t, Q_{1t}, \dots, Q_{nt})\}_{t=1}^T$ is consistent with the hypothesis that $(Q_{1t}, \dots, Q_{nt})$ maximizes *industry* profit at t for $t = 1, \dots, T$. In particular, show it is consistent with this collusion hypothesis for a sequence of inverse demands of the form $\tilde{P}_t(Q) = \max\{0, \alpha_t - \beta_t Q\}$ for $t = 1, \dots, T$, where each α_t and β_t are positive.

4. Consider the following modification of the well-known "Selten's horse" game:



A payoff vector of (x, y, z) denotes from left-to-right payoffs for players 1, 2, and 3. Answer the following questions regarding this game:

- (a) List all subgames.
- (b) Compute the set of all subgame perfect Nash equilibria.
- (c) Compute the set of sequential equilibria. For any strategy profile that you claim is part of a sequential equilibrium, be sure to verify that *all* criteria defining a sequential equilibrium are satisfied.

5. A monopolist serves a consumer who is one of two types θ_H, θ_L , with $P(\theta = \theta_H) = 1/2$. The monopolist's production costs are given by $c(q) = q^2/2$ and type θ 's utility over quantity-transfer pair (q, t) is given by $u(q, t, \theta) = \theta \cdot q - t$.

- (a) Assuming the consumer's type is publicly observable, solve for the first-best menu of contracts.
- (b) Now assuming types are private information, solve for the monopolist's optimal menu of contracts. Make the assumption that $2\theta_L > \theta_H > \theta_L$.